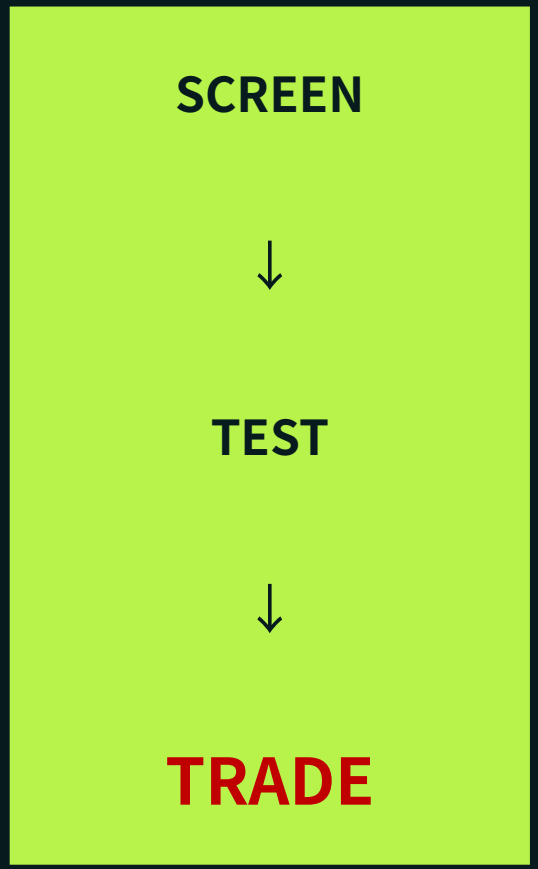
Five-Asset Cointegration Hand-Calculation Workbook

Screen → EG pair tests → Johansen rank → signal



Advanced Financial Time Series · Week 7

Find a tradable spread among five assets

Hypothetical — daily closes, T = 250; test statistics and eigenvalues assumed from software

Asset	Description
A	Refiner 1
B	Refiner 2
C	Crude-oil futures ETF
D	Crack-spread index
E	Spread between two rates

Order of work

- ① Step 0: order of integration of all five
- ② EG: test the rest pair by pair
- ③ EG step 3: half-life of passing pairs
- ④ Johansen: rank of the remaining set
- ⑤ Spread and today's trading signal

All you need: add · multiply · compare

Step 0: E is already stationary, so drop it; keep A–D

ADF t-stats, with constant — below the 5% value −2.86 rejects the 'unit root' null

Asset	Level ADF t	Diff. ADF t	Decision
A	−1.42	−15.3	**I(1)**
B	−1.87	−14.8	**I(1)**
C	−2.10	−16.1	**I(1)**
D	−0.95	−13.9	**I(1)**
E	−4.12	—	**I(0), dropped**

Decide by hand

- A: −1.42 > −2.86 → levels not rejected
- A: −15.3 < −2.86 → differences rejected → I(1)
- B, C, D: I(1) the same way
- E: −4.12 < −2.86 → stationary in levels
- E is no cointegration candidate — use it as is

Five in; four (A, B, C, D) go to the next step

EG step 2: of six pairs, only A–B and A–D are cointegrated

ADF on each pair's levels-regression residuals — EG 5% value −3.34 (2 variables)

Pair	Residual ADF t	Ordinary ADF −2.86	EG −3.34
A–B	−3.71	Pass	**Cointegrated**
A–C	−2.95	Pass	**No**
A–D	−3.52	Pass	**Cointegrated**
B–C	−2.40	Out	**No**
B–D	−3.05	Pass	**No**
C–D	−1.80	Out	**No**

Decide by hand

- −3.71 < −3.34 → A–B cointegrated
- −3.52 < −3.34 → A–D cointegrated
- A–C (−2.95), B–D (−3.05): pass ordinary ADF, fail EG
- Six pair tests: beware lucky passes

The two red rows are false cointegration from ordinary critical values

EG steps 1 and 3: A–B's spread today and its half-life

Take step-1 OLS (intercept 10, hedge ratio 0.95) and ECM speed −0.10 as given

Input	Value
Intercept	10
Hedge ratio	0.95
A today	100
B today	95
ECM adjustment speed	−0.10

Step 1: today's values into the spread (residual)

$$\hat{u}_t = A_t - 10 - 0.95\,B_t$$
$$= 100 - 10 - 0.95 \times 95 = -0.25$$

Step 3: adjustment speed → half-life

$$\rho = 1 + \hat{\alpha} = 1 - 0.10 = 0.90$$
$$h_{1/2} = \frac{\ln 0.5}{\ln 0.90} = \frac{-0.693}{-0.105} \approx 6.6 \text{ days}$$

Today's gap ≈ 0; if it widens, half closes in about 6.6 days — step 3 passed

Johansen: all four assets at once give four eigenvalues

Inputs A, B, C, D (all I(1)), lag p = 2 → k_ar_diff = 1, with constant · eigenvalues tabulated

i	Eigenvalue λ	−ln(1 − λ)
1	0.150	**0.1625**
2	0.060	**0.0619**
3	0.020	**0.0202**
4	0.008	**0.0080**

※ Right column precomputed on a calculator — by hand, use only this column

5% critical values (by k − r)

- k − r = 4 → trace 47.85, max-eigenvalue 27.59
- k − r = 3 → trace 29.80, max-eigenvalue 21.13
- k − r = 2 → trace 15.49, max-eigenvalue 14.26
- k − r = 1 → trace 3.84, max-eigenvalue 3.84

With k = 4, the critical value is 47.85 for r = 0, 29.80 for r = 1

Trace test: r = 0 rejected, r ≤ 1 not, so the rank is 1

Up from r = 0 — add the lower values of the right column, times T = 250

Null	trace	5% critical value	Decision
r = 0	63.2	47.85	**Reject → go on**
r ≤ 1	22.5	29.80	**Stop**
r ≤ 2	7.1	15.49	**(check)**
r ≤ 3	2.0	3.84	**(check)**

r = 0: add all four values

$$\lambda_{\text{trace}}(0) = 250 \times (0.1625 + 0.0619 + 0.0202 + 0.0080)$$
$$= 250 \times 0.2526 = 63.2 \; > \; 47.85$$

r ≤ 1: all but the first

$$\lambda_{\text{trace}}(1) = 250 \times (0.0619 + 0.0202 + 0.0080)$$
$$= 250 \times 0.0901 = 22.5 \; < \; 29.80$$

First non-rejection at r = 1 — one independent long-run equilibrium among the four

The max-eigenvalue test gives the same answer: rank 1

One eigenvalue at a time — exactly r vs r + 1

Null	Max-eigenvalue	5% critical value	Decision
r = 0	40.6	27.59	**Reject**
r = 1	15.5	21.13	**Stop**

※ If the tests disagree, re-check with another sample period or lag

250 × −ln(1 − λ), one by one

$$\lambda_{\max}(0) = 250 \times 0.1625 = 40.6 \;>\; 27.59$$
$$\lambda_{\max}(1) = 250 \times 0.0619 = 15.5 \;<\; 21.13$$

Trace and max-eigenvalue both say rank 1 — a robust result

The spread from the cointegrating vector gives a sell signal today

First vector normalised on A (1, −0.62, −0.25, −0.10) · past spread mean 20.0, s.d. 1.0

Asset	Today's price	Vector coefficient
A	100	**1**
B	95	**−0.62**
C	60	**−0.25**
D	40	**−0.10**

Spread = sum of price × coefficient

$$z = A - 0.62\,B - 0.25\,C - 0.10\,D$$
$$= 100 - 58.9 - 15.0 - 4.0 = 22.1$$

z-score and entry rule

$$Z = \frac{z - \mu_z}{\sigma_z} = \frac{22.1 - 20.0}{1.0} = 2.1 \geq 2$$

Z = 2.1 ≥ 2 → sell 1 unit of A, buy B 0.62 · C 0.25 · D 0.10; exit when Z is near 0

From five assets down to one spread

At each step, compare one number with a cut-off to set the next move

Step	Number compared	Conclusion
Step 0	**E: −4.12 < −2.86**	**Drop E, keep A–D**
EG step 2	**A–B −3.71, A–D −3.52 < −3.34**	**Only two pairs**
EG step 3	**Half-life 6.6 days**	**A–B tradable**
Johansen	**63.2 > 47.85, 22.5 < 29.80**	**Rank 1**
Signal	**Z = 2.1 ≥ 2**	**Enter: sell the spread**

All figures hypothetical — with real data, also check for structural breaks and sample period

Screen — step 0

Find pairs — Engle-Granger

See the system — Johansen

Next
Error correction and pairs trading (Part 3)